

MM-207

Centrifugal Pump Maintenance Manual

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1 Scope

This manual covers scheduled and condition-based maintenance of horizontal centrifugal pumps in boiler feed, cooling water and transfer duty. It defines inspection intervals, wear limits and the conditions under which a pump must be removed from service.

NOTE Where this manual and an operating procedure disagree on a mechanical intervention, this manual governs. Where this manual and a safety instruction disagree, the safety instruction governs.

6 Condition Monitoring

6.1 Vibration

Overall vibration velocity shall be trended monthly at the inboard and outboard bearing housings. An increase of more than 40 percent over the established baseline warrants investigation regardless of absolute value.

6.2 Bearing Temperature

Bearing temperature is trended continuously. A sustained rise of more than 15 K above the established operating band, or any excursion above 95 degrees Celsius, requires investigation within one shift.

Bearing temperature that rises after a suction-side event and returns to band once suction pressure recovers indicates a consequential thermal excursion rather than a primary bearing defect. Repeated consequential excursions nonetheless accumulate damage and are counted for the purposes of §7.3.

7 Inspection Intervals

7.1 Scheduled Intervals

Component	Interval	Basis
Mechanical seal faces	Annual	Calendar
Inboard bearing	Annual	Calendar
Outboard bearing	Annual	Calendar
Coupling alignment	24 months	Calendar
Impeller clearance	At overhaul	Condition

Table 7-1 — Baseline inspection intervals

7.2 Deferral

A scheduled inspection may be deferred by up to 90 days on the authority of the responsible reliability engineer, provided the conditions in §7.3 are not met and condition monitoring shows no adverse trend.

7.3 Mandatory Removal from Service

A pump exhibiting more than five cavitation-induced suction transients within any rolling 30-day period shall be removed from service at the next scheduled shutdown for inboard bearing and mechanical seal inspection. Such a pump shall not be returned to extended service on the basis of increased monitoring alone.

WARNING This requirement is not satisfied by trending, by increased inspection frequency, or by a recommendation to continue operation issued by a condition-monitoring or advisory system. Removal from service at the next scheduled shutdown is mandatory once the threshold is met.

Sustained operation below 40 percent of best efficiency point accelerates thrust bearing wear. Where recurring suction transients are recorded, inspect the inboard bearing and mechanical seal faces at the next available outage rather than deferring to the annual interval under §7.1.

7.4 Post-Inspection Criteria

Measurement	Acceptable	Replace
Bearing radial clearance	Below 0.12 mm	0.12 mm or above
Seal face flatness	Within 2 helium light bands	Above 2 bands
Shaft runout at seal	Below 0.05 mm	0.05 mm or above
Impeller wear ring clearance	Below 0.75 mm	0.75 mm or above

Table 7-4 — Wear limits

8 Spares and Consumables

Hold one complete seal cartridge and one bearing set per pump model on site. Lead time for the Sulzer HPT series cartridge is 14 weeks; a pump meeting the §7.3 criteria without a cartridge on hand is a reportable

spares risk.